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Q1. CASE STUDY 1

CODE

```
#include <stdio.h>
#include <string.h>


union EmploymentStatus {
    char fullTime[15];
    char contract[15];
    char intern[15];
};


struct Project {
    char projectName[30];
    char role[20];
    int durationMonths;
};


struct Employee {
    int empID;
    char name[30];
    char department[20];
    char designation[20];
    float salary;
    struct Project project;
    union EmploymentStatus status;
    int statusType;
};


static int totalEmployees = 0;

struct Employee empRecords[100];

void addEmployee(int id, char *name, char *dept, char *desig, float sal,
                  char *projName, char *role, int duration,
                  int statusType, char *statusValue) {

    struct Employee emp;
    emp.empID = id;
    strcpy(emp.name, name);
    strcpy(emp.department, dept);
    strcpy(emp.designation, desig);
    emp.salary = sal;


    strcpy(emp.project.projectName, projName);
    strcpy(emp.project.role, role);
    emp.project.durationMonths = duration;


    emp.statusType = statusType;
    if (statusType == 0)
        strcpy(emp.status.fullTime, statusValue);
    else if (statusType == 1)
        strcpy(emp.status.contract, statusValue);
    else
        strcpy(emp.status.intern, statusValue);

    empRecords[totalEmployees] = emp;
    totalEmployees++;
}
```

```
void displayEmployee(struct Employee emp) {
    printf("\nEmployee ID: %d\n", emp.empID);
    printf("Name: %s\n", emp.name);
    printf("Department: %s\n", emp.department);
    printf("Designation: %s\n", emp.designation);
    printf("Salary: %.2f\n", emp.salary);
    printf("Project: %s | Role: %s | Duration: %d months\n",
           emp.project.projectName, emp.project.role, emp.project.durationMonths);

    printf("Employment Status: ");
    if (emp.statusType == 0)
        printf("%s (Full-Time)\n", emp.status.fullTime);
    else if (emp.statusType == 1)
        printf("%s (Contract)\n", emp.status.contract);
    else
        printf("%s (Intern)\n", emp.status.intern);
}

int main() {

    addEmployee(101, "Arun Kumar", "IT", "Software Engineer", 55000.00,
                "Cloud HRMS", "Backend Developer", 12, 0, "Full-Time");

    addEmployee(102, "Priya Sharma", "IT", "Data Analyst", 40000.00,
                "AI Analytics", "Data Analyst", 6, 1, "Contract");

    addEmployee(103, "Ravi Teja", "IT", "Intern Developer", 15000.00,
                "Mobile App", "Frontend Intern", 3, 2, "Intern");

    printf("===== Employee Records =====\n");
    for (int i = 0; i < totalEmployees; i++) {
        displayEmployee(empRecords[i]);
    }

    printf("\nTotal Employees (Global Count using static storage class): %d\n", totalEmployees);

    return 0;
}
```

OUTPUT

```
(no output)
```

Q2. CASE STUDY 2

CODE

```
#include <stdio.h>
#include <string.h>

struct Tester {
    int testerID;
    char testerName[50];
    char testerContact[15];
};

struct Developer {
    int devID;
    char devName[50];
    char devContact[15];
};


union BugStatus {
    char open[10];
    char inProgress[15];
    char closed[10];
};


struct Bug {
    int bugID;
    char description[100];
    char severity[20];
    struct Developer dev;
    struct Tester tester;
    union BugStatus status;
    int statusCode;
};

void displayBug(struct Bug b) {
    printf("\n————— Bug Report —————\n");
    printf("Bug ID          : %d\n", b.bugID);
    printf("Description     : %s\n", b.description);
    printf("Severity        : %s\n", b.severity);

    printf("\n-- Developer Details --\n");
    printf("Developer ID    : %d\n", b.dev.devID);
    printf("Developer Name  : %s\n", b.dev.devName);
    printf("Developer Phone : %s\n", b.dev.devContact);

    printf("\n-- Tester Details --\n");
    printf("Tester ID       : %d\n", b.tester.testerID);
    printf("Tester Name     : %s\n", b.tester.testerName);
    printf("Tester Phone    : %s\n", b.tester.testerContact);

    printf("\nCurrent Status  : ");
    if (b.statusCode == 1)
        printf("%s\n", b.status.open);
    else if (b.statusCode == 2)
        printf("%s\n", b.status.inProgress);
    else if (b.statusCode == 3)
        printf("%s\n", b.status.closed);
    else
        printf("Unknown\n");
    printf("————————————————————————\n");
}

int main() {
```

```
    struct Bug bug1;
    int choice;

    bug1.bugID = 101;
    strcpy(bug1.description, "Login page crashes on invalid input");
    strcpy(bug1.severity, "High");

    bug1.dev.devID = 201;
    strcpy(bug1.dev.devName, "Rahul Sharma");
    strcpy(bug1.dev.devContact, "9876543210");


    bug1.tester.testerID = 301;
    strcpy(bug1.tester.testerName, "Priya Verma");
    strcpy(bug1.tester.testerContact, "9123456780");


    printf("Enter status (1-Open, 2-In Progress, 3-Closed): ");
    scanf("%d", &choice);

    bug1.statusCode = choice;
    if (choice == 1)
        strcpy(bug1.status.open, "Open");
    else if (choice == 2)
        strcpy(bug1.status.inProgress, "In Progress");
    else if (choice == 3)
        strcpy(bug1.status.closed, "Closed");
    else
        printf("Invalid status entered.\n");


    displayBug(bug1);

    return 0;
}
```

INPUT

```
2
```

OUTPUT

```
(no output)
```

Q3. CASE STUDY 3

CODE

```
#include <stdio.h>
#include <string.h>


struct Supplier {
    int supplierID;
    char supplierName[50];
    char contactNumber[15];
};


union CategoryDetails {
    struct {
        char brand[30];
        int warrantyMonths;
    } electronics;

    struct {
        char size[10];
        char material[30];
    } clothing;

    struct {
        char expiryDate[15];
    } grocery;
};

enum ProductType { ELECTRONICS, CLOTHING, GROCERY };

struct Product {
    int productID;
    char productName[50];
    float price;
    int stockQuantity;
    enum ProductType type;
    union CategoryDetails categoryInfo;
    struct Supplier supplierInfo;
};

void displayProduct(struct Product p) {
    printf("\n--- Product Details ---\n");
    printf("Product ID       : %d\n", p.productID);
    printf("Product Name     : %s\n", p.productName);
    printf("Price            : %.2f\n", p.price);
    printf("Stock Quantity   : %d\n", p.stockQuantity);

    printf("Category         : ");
    if (p.type == ELECTRONICS) {
        printf("Electronics\n");
        printf("Brand            : %s\n", p.categoryInfo.electronics.brand);
        printf("Warranty (Months): %d\n", p.categoryInfo.electronics.warrantyMonths);
    } else if (p.type == CLOTHING) {
        printf("Clothing\n");
        printf("Size             : %s\n", p.categoryInfo.clothing.size);
        printf("Material         : %s\n", p.categoryInfo.clothing.material);
    } else if (p.type == GROCERY) {
        printf("Grocery\n");
        printf("Expiry Date      : %s\n", p.categoryInfo.grocery.expiryDate);
    }

    printf("\n--- Supplier Details ---\n");
    printf("Supplier ID      : %d\n", p.supplierInfo.supplierID);
```

```
    printf("Supplier Name    : %s\n", p.supplierInfo.supplierName);
    printf("Contact Number   : %s\n", p.supplierInfo.contactNumber);
}

int main() {
    struct Product p1;

    p1.productID = 101;
    strcpy(p1.productName, "Smartphone");
    p1.price = 24999.50;
    p1.stockQuantity = 50;

    p1.type = ELECTRONICS;
    strcpy(p1.categoryInfo.electronics.brand, "TechBrand");
    p1.categoryInfo.electronics.warrantyMonths = 12;

    p1.supplierInfo.supplierID = 501;
    strcpy(p1.supplierInfo.supplierName, "Global Electronics Ltd");
    strcpy(p1.supplierInfo.contactNumber, "9876543210");

    displayProduct(p1);

    struct Product p2;
    p2.productID = 102;
    strcpy(p2.productName, "T-Shirt");
    p2.price = 499.00;
    p2.stockQuantity = 200;

    p2.type = CLOTHING;
    strcpy(p2.categoryInfo.clothing.size, "L");
    strcpy(p2.categoryInfo.clothing.material, "Cotton");

    p2.supplierInfo.supplierID = 502;
    strcpy(p2.supplierInfo.supplierName, "Fashion Hub");
    strcpy(p2.supplierInfo.contactNumber, "9123456789");

    displayProduct(p2);

    return 0;
}
```

OUTPUT

```
(no output)
```

Q4. CASE STUDY 4

CODE

```
#include <stdio.h>
#include <string.h>

union PatientStatus {
    char admitted[20];
    char discharged[20];
    char observation[20];
};

struct Visit {
    char diagnosis[50];
    char doctorName[50];
    float treatmentCost;
};

struct Patient {
    int patientID;
    char name[50];
    int age;
    char bloodGroup[5];
    char medicalHistory[100];
    struct Visit visitDetails;
    union PatientStatus status;
};

int main() {
    struct Patient p;
    int statusChoice;


    printf("Enter Patient ID: ");
    scanf("%d", &p.patientID);
    getchar();

    printf("Enter Name: ");
    fgets(p.name, sizeof(p.name), stdin);
    p.name[strcspn(p.name, "\n")] = 0;

    printf("Enter Age: ");
    scanf("%d", &p.age);
    getchar();

    printf("Enter Blood Group: ");
    scanf("%s", p.bloodGroup);
    getchar();

    printf("Enter Medical History: ");
    fgets(p.medicalHistory, sizeof(p.medicalHistory), stdin);
    p.medicalHistory[strcspn(p.medicalHistory, "\n")] = 0;


    printf("Enter Diagnosis: ");
    fgets(p.visitDetails.diagnosis, sizeof(p.visitDetails.diagnosis), stdin);
    p.visitDetails.diagnosis[strcspn(p.visitDetails.diagnosis, "\n")] = 0;

    printf("Enter Doctor Name: ");
    fgets(p.visitDetails.doctorName, sizeof(p.visitDetails.doctorName), stdin);
    p.visitDetails.doctorName[strcspn(p.visitDetails.doctorName, "\n")] = 0;

    printf("Enter Treatment Cost: ");
    scanf("%f", &p.visitDetails.treatmentCost);
    getchar();
```

```
    printf("\nSelect Status (1. Admitted, 2. Discharged, 3. Under Observation): ");
    scanf("%d", &statusChoice);
    getchar();

    if (statusChoice == 1) {
        strcpy(p.status.admitted, "Admitted");
    } else if (statusChoice == 2) {
        strcpy(p.status.discharged, "Discharged");
    } else {
        strcpy(p.status.observation, "Under Observation");
    }


    printf("\n--- Patient Record ---\n");
    printf("ID: %d\n", p.patientID);
    printf("Name: %s\n", p.name);
    printf("Age: %d\n", p.age);
    printf("Blood Group: %s\n", p.bloodGroup);
    printf("History: %s\n", p.medicalHistory);
    printf("Status: %s\n", p.status.admitted);

    printf("\n--- Visit Details ---\n");
    printf("Diagnosis: %s\n", p.visitDetails.diagnosis);
    printf("Doctor: %s\n", p.visitDetails.doctorName);
    printf("Cost: %.2f\n", p.visitDetails.treatmentCost);

    return 0;
}
```

OUTPUT

```
(no output)
```

Student 32 **Srikanth S** 192514055RB 0 / 4 submitted

Q1. CASE STUDY 1

— Not Submitted —

Q2. CASE STUDY 2

— Not Submitted —

Q3. CASE STUDY 3

— Not Submitted —

Q4. CASE STUDY 4

— Not Submitted —